UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: ARTIFICIAL INTELLEGENCE

SUBMITTED BY & TO

NAME : RIFAQ AJMAL

REG NO : 23MDBCS 435

SECTION : A

SEMESTER : 4th

SUBMITTED T0 : MAM FAIZA

LAB # 4

EXAMPLE 1.

Write a program that read the temperature from a text file in the folder, and convert that from celcius to farnhite.

TEMPERATURE IN CELCIUS

temps - Notepad

File Edit Format View Help

```
-10.0
0.0
5.0
10.0
15.0
20.0
25.0
30.0
35.0
40.0
```

PROGRAM

1st example.py - C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\1st example.py (3.13.2)

File Edit Format Run Options Window Help

```
file1 = open('ftemps.txt', 'w')
temperatures = [line.strip() for line in open('temps.txt')]

for t in temperatures:
    # Convert the string temperature to a float
    t = float(t)
    # Convert Celsius to Fahrenheit and write to the file
    print(t * 9/5 + 32, file=file1)

file1.close()

with open('ftemps.txt', 'r') as ftemps:
    print("Temperatures in Fahrenheit:")
    for line in ftemps:
        print(line.strip())
```

OUTPUT

TEMPERATUTE CONVERTED TO FARNHITE

```
IDLE Shell 3.13.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\1st example.py =
Temperatures in Fahrenheit:
14.0
32.0
41.0
50.0
59.0
68.0
77.0
86.0
95.0
104.0
```

EXAMPLE 2

Write a program which ask questions from you and when you give the answer so that will be checked wit the file in folder.

If the answer is correct will show correct else incorrect. And in the last it will give you a score according to your answers.

FILE WITH WHICH THE ANSWERS SHOULD BE CHECK.

```
answers - Notepad
File  Edit  Format  View  Help
Paris
4
Shakespeare
```

PRORGAM

2nd examples.py - C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\2nd examples.py (3.13.2)

File Edit Format Run Options Window Help

```python
# Read questions and answers from files
questions = [line.strip() for line in open('questions.txt')]
answers = [line.strip() for line in open('answers.txt')]

score = 0  # Keep track of the score
total_questions = len(questions)

# Ask each question
for i in range(total_questions):
    print(f"Question {i + 1}: {questions[i]}")
    user_answer = input("Your answer: ").strip()

    # Check if the answer is correct
    if user_answer.lower() == answers[i].lower():
        print("Correct!\n")
        score += 1
    else:
        print(f"Wrong. The correct answer is: {answers[i]}\n")

# Show final score
print(f"You got {score} out of {total_questions} questions right.")
```

OUTPUT

IDLE Shell 3.13.2

File Edit Shell Debug Options Window Help

```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\2nd example
Question 1: What is the capital of France?
Your answer: Paris
Correct!

Question 2: What is the square root of 16?
Your answer: 4
Correct!

Question 3: Who wrote 'Hamlet'?
Your answer: shakesspare
Wrong. The correct answer is: Shakespeare

You got 2 out of 3 questions right.
>>>
```

EXAMPLE 3

Write a program that Read the file from the folder and split each line into a list of game information.

Initialize variables to keep track of the biggest difference and game info.

Score of the first team and Score of the second team and Calculate the margin of victory.

Output the game information with the biggest margin of victory.

FILE IN THE FOLDER WHICH HAVE THE INFORMATION

scores - Notepad

File Edit Format View Help

```
02/27/2010, Robert Morris, 61, Mount St. Mary's, 63
12/03/2009, Salem International, 35, Marshall, 119
01/15/2010, Duke, 75, North Carolina, 60
12/22/2009, Kansas, 102, Dayton, 55
11/10/2009, Louisville, 91, Western Kentucky, 54
03/08/2010, Michigan State, 67, Purdue, 63
02/14/2010, Syracuse, 89, Georgetown, 85
01/02/2010, Kentucky, 85, Tennessee, 80
```

PROGRAM

3rd exampless.py - C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\3rd exampless.py (3.13.2)

File Edit Format Run Options Window Help

```python
# Read the file and split each line into a list of game information
lines = [line.strip() for line in open('scores.txt')]
games = [line.split(',') for line in lines]

# Initialize variables to keep track of the biggest difference and game info
biggest_diff = 0
game_info = []

# Loop through each game to find the biggest margin of victory
for g in games:
    team1_score = int(g[2])  # Score of the first team
    team2_score = int(g[4])  # Score of the second team
    diff = abs(team1_score - team2_score)  # Calculate the margin of victory

    if diff > biggest_diff:  # If current margin is larger, update biggest_diff
        biggest_diff = diff
        game_info = g  # Store the game information

# Output the game information with the biggest margin of victory
print(f"Most lopsided game: {game_info}")
print(f"Date: {game_info[0]}")
print(f"Teams: {game_info[1]} vs {game_info[3]}")
print(f"Score: {game_info[1]} {game_info[2]} - {game_info[3]} {game_info[4]}")
print(f"Margin of Victory: {biggest_diff}")
```

OUTPUT

IDLE Shell 3.13.2

File Edit Shell Debug Options Window Help

```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\3rd exampless.py
Most lopsided game: ['12/03/2009', ' Salem International', ' 35', ' Marshall', ' 119']
Date: 12/03/2009
Teams:  Salem International vs  Marshall
Score:  Salem International  35 -  Marshall  119
Margin of Victory: 84
>>>
```

EXAMPLE 4

Write a program that Load the paragraph from the file and split it into word, Use regex to split text into words, ignoring punctuation.

Example 1: Print all three-letter words

Example 2: Print all the words that start with 'gn' or 'kn'

Example 3: Determine what percentage of words start with a vowel

Example 4: Print all 7-letter words that start with 'th' and end with 'ly'

Example 5: Print the first ten words that start with 'q'

Example 6: Find the longest word that can be made using only the letters a, b, c, d, and e.

FILE FROM WHICH THE PROGRAM WILL READ.

*wordlist - Notepad

File Edit Format View Help

```
Exploring and manipulating a wordlist in Python allows us to perform a variety of
useful operations on words. By iterating through the list, we can filter words based
on their length, specific letter patterns, or prefixes and suffixes. For example, we
can quickly print all three-letter words or find words that start with specific  letter
combinations like "gn" or "kn". Python also provides us with the ability to calculate
statistics from the list, such as determining what percentage of words start with vowels.
Additionally, searching for specific word  patterns, such as seven-letter words starting
with "th" and ending with "ly", can be very helpful for solving word puzzles. If we need
to focus on a certain letter, like 'q', we can efficiently extract the first ten words
starting with that letter. Lastly, using loops and conditions, we can even find the longest
word composed only of a limited set of letters, such as 'a', 'b', 'c', 'd', and 'e'. These
examples demonstrate the power and flexibility of Python when working with large wordlists,
making it an excellent tool for word games, linguistics research, and text analysis.

watch
banana
fast
king
education
gnome
thisely
abccdee
thelady
thorely
abcd
```

PRORGAM

4th wordplay.py - C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\4th wordplay.py (3.13.2)

File Edit Format Run Options Window Help

```python
import re

# Load the paragraph from the file and split it into words
with open('wordlist.txt', 'r') as file:
    text = file.read()

# Use regex to split text into words, ignoring punctuation
wordlist = re.findall(r'\b\w+\b', text)

# Example 1: Print all three-letter words
print("Three-letter words:")
for word in wordlist:
    if len(word) == 3:
        print(word)

# Example 2: Print all the words that start with 'gn' or 'kn'
print("\nWords that start with 'gn' or 'kn':")
for word in wordlist:
    if word[:2] == 'gn' or word[:2] == 'kn':
        print(word)
```

```python
# Example 3: Determine what percentage of words start with a vowel
count = 0
for word in wordlist:
    if word[0].lower() in 'aeiou':
        count += 1
percentage = 100 * count / len(wordlist)
print(f"\nPercentage of words that start with a vowel: {percentage:.2f}%")

# Example 4: Print all 7-letter words that start with 'th' and end with 'ly'
print("\n7-letter words that start with 'th' and end with 'ly':")
for word in wordlist:
    if len(word) == 7 and word[:2] == 'th' and word[-2:] == 'ly':
        print(word)

# Example 5: Print the first ten words that start with 'q'
print("\nFirst ten words that start with 'q':")
q_words = [word for word in wordlist if word[0].lower() == 'q']
print(q_words[:10])

# Example 6: Find the longest word that can be made using only the letters a, b, c, d, and e
largest = 0
largest_word = ''
for word in wordlist:
    for c in word:
        if c not in 'abcde':
            break
    else:
        if len(word) > largest:
            largest = len(word)
            largest_word = word
print(f"\nLongest word using only 'a', 'b', 'c', 'd', 'e': {largest_word}")
```

OUTPUT

IDLE Shell 3.13.2

File Edit Shell Debug Options Window Help

```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Rifaq Ajmal\Desktop\lab 4 Practice Examples\4th wordplay.py
Three-letter words:
and
the
can
and
For
can
all
the
the
for
and
can
for
can
the
ten
and
can
the
set
and
the
and
for
and
```

```
Words that start with 'gn' or 'kn':
gn
kn
knowledge
gnome

Percentage of words that start with a vowel: 27.64%

7-letter words that start with 'th' and end with 'ly':
thisely
thorely

First ten words that start with 'q':
['quickly', 'q', 'quick', 'queen']

Longest word using only 'a', 'b', 'c', 'd', 'e': abccdee
>>>
```

THE END